

IARSLCE 2026 · PRE-CONFERENCE WORKSHOP

Participatory AI for Community Engagement

Nicholas Mattei & Saad Hassan

Tulane Center for Community-Engaged Artificial Intelligence

Sunday, October 11, 2026 · 9:00 – 11:30 am
Monroe Hall 152, Tulane University, New Orleans



Tulane University
Center for Community-Engaged
Artificial Intelligence

Today's plan

🕒 9:00 – 9:10 · 10 min

9:00	Welcome and introductions	Nick & Saad
9:10	What is AI, and where are we?	Nick
9:25	Designing with, not for	Saad
9:40	Reflect: your one problem	Everyone
9:55	Share-outs	Everyone
10:10	Break	We cluster your ideas
10:25	Breakout groups: three rounds	Two groups
11:10	Report back	Each group
11:20	Debrief and next steps	Everyone



Slides, readings, and links

saadh.info/pace2

Introductions

🕒 9:00 – 9:10 · 10 min



nickmattei.net

Nicholas Mattei

Associate Professor of Computer Science, Tulane University
Co-Director, Tulane Center for Community-Engaged AI

- AI, machine learning, and decision making with real-world stakeholders
- Community projects across New Orleans: civic tech, courts, utilities
- Co-author, Computing and Technology Ethics: Engaging Through Science Fiction (MIT Press)

Chair, ACM SIGAI

NSF CAREER

Prev. NASA · IBM · UNSW

Introductions

🕒 9:00 – 9:10 · 10 min



saadh.info

Saad Hassan

Assistant Professor of Computer Science, Tulane University
Director, NOLA A11y Lab · Affiliate, CEAI

- Human–AI systems that make communication, information, and public services accessible
- Sign language and captioning technologies with Deaf and hard-of-hearing communities
- Civic AI and accessibility dashboards with community organizations

PhD, RIT

Visiting: Google & Meta

CHI · ASSETS · NeurIPS

Let's get to know you

🕒 9:00 – 9:10 · 10 min

1

Your name

and how you'd like to be called

2

Your affiliation

institution, office, or organization

3

**One community partner
or project**

you're working with right now

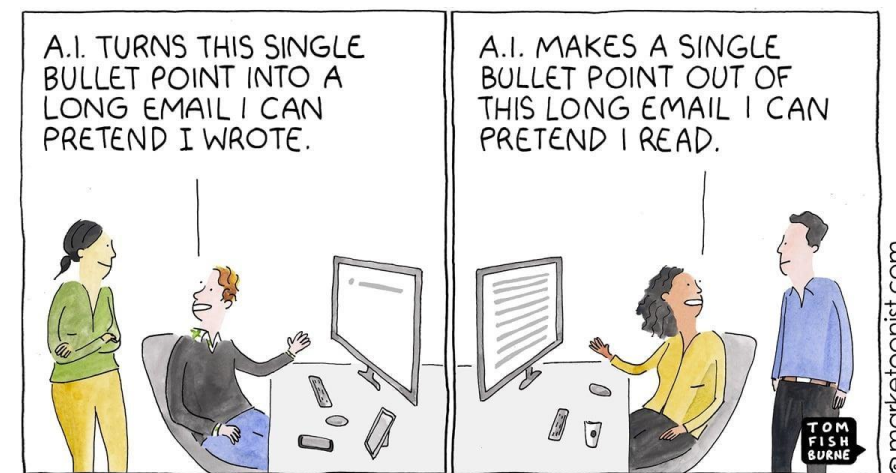
About 30 seconds each

The pre-workshop survey covers the rest, so keep it short.

PART 1

What is AI, and where are we?

Nicholas Mattei



AI is already everywhere

🕒 9:10 – 9:25 · 15 min

Feeds, search, prices, matches, rides, loans: most of it has run on AI for years.



Everyone is a programmer now

🕒 9:10 – 9:25 · 15 min

THE GOAL

Make intent clear to computers!

Wires

>

Punch cards

>

Languages

>

No-code

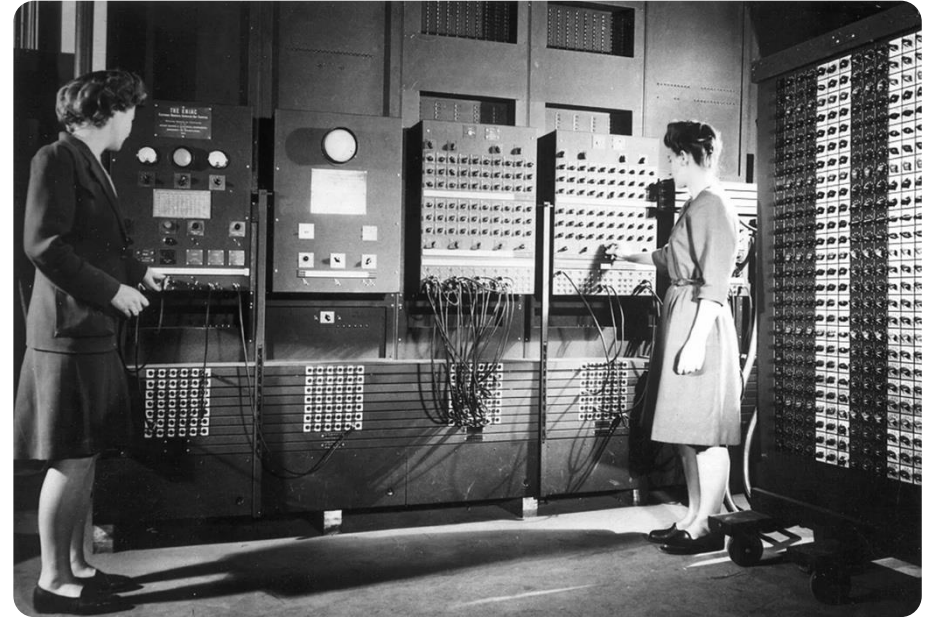
>

Prompt as code

CS folks should **embrace this!**

The magic of CS is that we get to make the machine do what we want.
Now everyone can!

Secondary argument: we've been in the bot age for a while (search, etc. is the product). People never produced this... (well, in the 90s maybe).



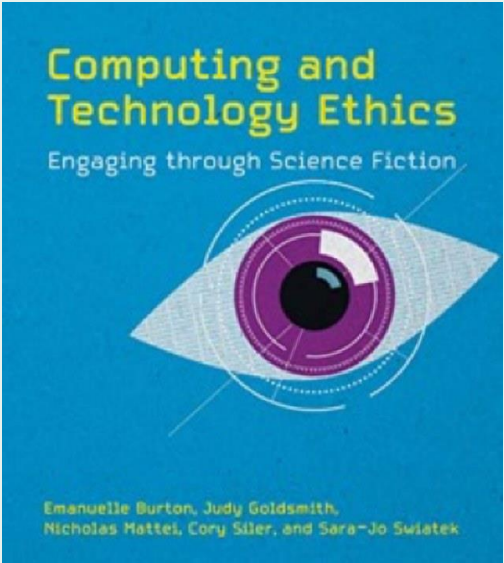
READ MORE

[The End of Programming as We Know It](#) (O'Reilly)

Teaching computing and technology ethics

🕒 9:10 – 9:25 · 15 min

Engaging through science fiction



AVAILABLE NOW · MIT PRESS

CHAPTERS

Ethical Frameworks

Managing Knowledge

Privacy and Personhood

Technology and Society

Professional Ethics

ANTHOLOGY

Short stories from acclaimed writers including Ken Liu, Elizabeth Bear, Paolo Bacigalupi, and T. C. Boyle animate ethical challenges in computing technology.



GET THE BOOK

rebrand.ly/SciFi-Book

Let's step back: what is technology?

🕒 9:10 – 9:25 · 15 min

1

AI and technology is not a single thing or only objects.

It does not have agency, it is not a person, and by many philosophical/religious arguments cannot be “*ethical*.”

2

Focusing only on specific objects or artifacts: a camera, a traffic light, a gun, an app, a bike...

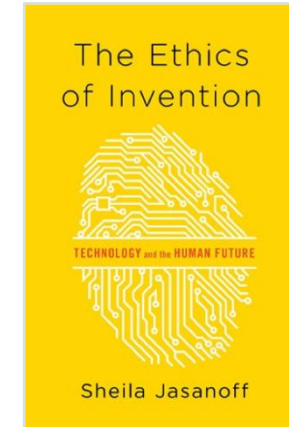
How technology affects society is an insufficient question of cause and effect: *does a gun do bad things?*

3

Technology and society are bound together in an inseparable way: our choices about which technologies to develop, or not to develop, often grow out of the society in which we live.

And those technologies, in turn, change our social settings and the subsequent technologies that are produced in the context in which we live.

Consider the stoplight...



Sheila Jasanoff
The Ethics of Invention



Consider the stoplight

🕒 9:10 – 9:25 · 15 min

INNOVATION

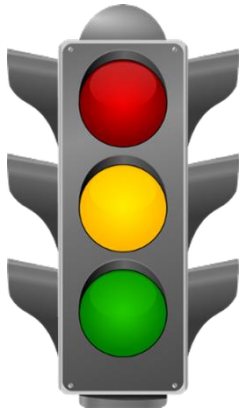
A Brief History of the Stoplight

How a bright idea shaped our cities and gave the go-ahead to our love affair with the car

Megan Kate Nelson

Author, *Saving Yellowstone: Exploration and Preservation in Reconstruction America*
May 2018

An additional complaint that gained traction was the device's unfortunate impact on civility. Long before today's epidemic of road rage, critics warned that drivers had surrendered some of their humanity; they didn't have to acknowledge each other or pedestrians at intersections, but rather just stare at the light and wait for it to change. As



THE FIRST SIGNAL

London, 1868. A gas-lit semaphore outside the Houses of Parliament.

the Houses of Parliament.^[8] The design combined three [semaphore](#) arms with red and green gas lamps for night-time use, on a pillar, operated by a [police constable](#). The gas lantern was manually turned by a traffic police officer with a lever at its base so that the appropriate light faced traffic.^[9] The signal was 22 feet (6.7 m) high. The light was called the semaphore and had arms that would extend horizontally that commanded drivers to "Stop" and then the arms would lower to a 45 degrees angle to tell drivers to proceed with "Caution". At night a red light would command "Stop" and a green light would mean use "Caution".^[4] Although it was said to be successful at controlling traffic, its operational life was brief. It exploded on 2 January 1869 as a result of a leak in one of the gas lines underneath the pavement^[10] and injured the policeman who was operating it.^[11]

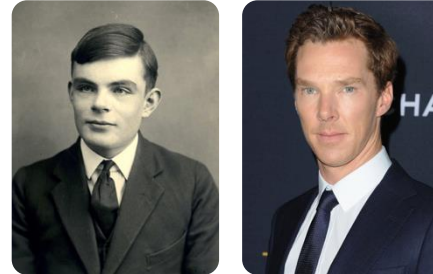
Sources: [Smithsonian Magazine](#) · [Wikipedia, History of traffic lights](#)

Wait, what is AI?

🕒 9:10 – 9:25 · 15 min

ALAN TURING, 1950

“I propose to consider the question, ‘Can machines think?’ ... I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.”



TODAY

Study finds ChatGPT's latest bot behaves like humans, only better

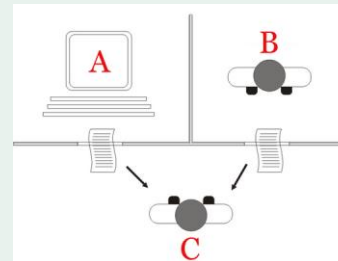
The most recent version of ChatGPT passes a rigorous Turing test, diverging from average human behavior chiefly to be more cooperative.

Passes a rigorous Turing test. Scan for the study.



THE IMITATION GAME

“... a man (A), a woman (B) and an interrogator (C) ... The object of the game for the interrogator is to determine which of the other two is the man and which is the woman.”

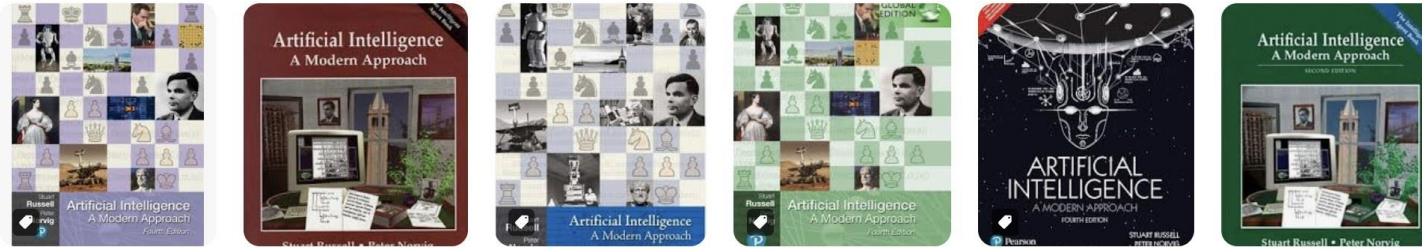


Tessler's Theorem: *“Intelligence is what machines haven't done yet.”*

I.e., “If machines can do it, it isn't intelligence.”

AI for AI researchers

🕒 9:10 – 9:25 · 15 min



“‘Artificial intelligence’ is when a machine mimics ‘cognitive functions’ that humans associate with other human minds such as ‘learning’ and ‘problem solving.’”

AGENT-BASED APPROACH

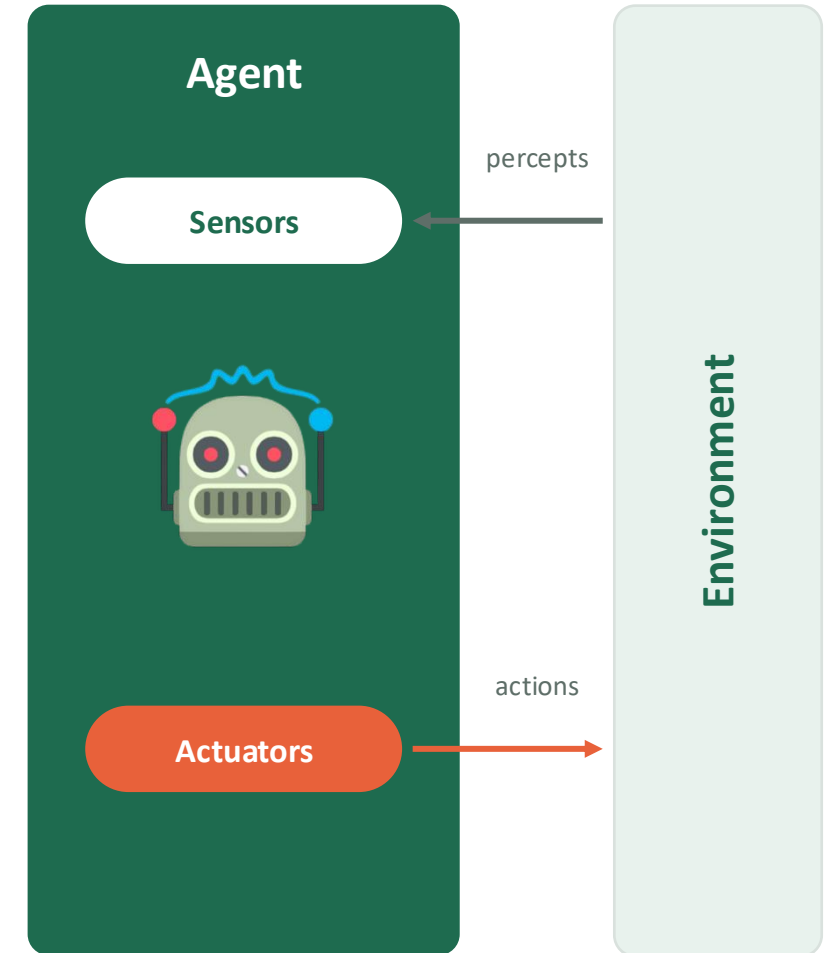
“... rational agents that act so as to achieve the best outcome, or, when there is uncertainty, the best expected outcome.”

TRADITIONAL GOALS

Reasoning, planning, and knowledge representation

Machine learning and natural language processing

Perception (vision) and embodiment (robotics)



Text-to-text systems

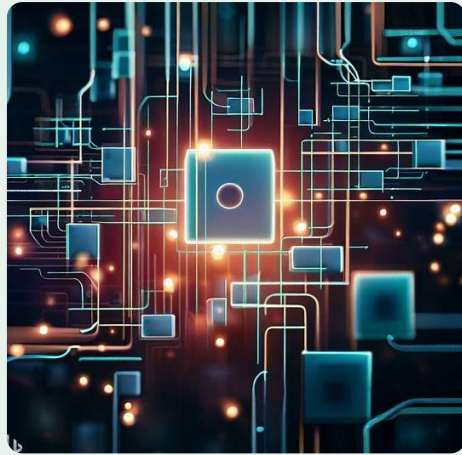
🕒 9:10 – 9:25 · 15 min



01

The data

Scrape LOTS of TEXT from the internet.



02

The model

Create a giant statistical model that relates how WORDS appear with other WORDS.



give me the
Gettysburg
address

03

The prompt

You type a request.



**Four score and
seven years ago
our forefathers
brought...**

04

The output

The model continues the text, one word at a time.

Given the words so far, keep predicting

🕒 9:10 – 9:25 · 15 min

The best thing about AI is its ability to,

The best thing about AI is its ability to **create**,

The best thing about AI is its ability to create **worlds**,

The best thing about AI is its ability to create worlds **that**,

The best thing about AI is its ability to create worlds that **are**,

The best thing about AI is its ability to create worlds that are **both**,

The best thing about AI is its ability to create worlds that are both **exciting**,

Source: Stephen Wolfram, “What Is ChatGPT Doing ... and Why Does It Work?” (2023)

General idea: predict the next word



The best thing about AI is its ability to ...

MOST LIKELY NEXT WORDS



Values from Stephen Wolfram, “What Is ChatGPT Doing ... and Why Does It Work?” (2023)

Three ways to tell a computer what to do

🕒 9:10 – 9:25 · 15 min

1

Traditional programming

The computer does exactly what you tell it, whether you meant it or not.

Rules written by people

2

Machine learning

Learns patterns from data to predict, classify, and recommend.

Rules learned from data

3

Generative AI

Trained broadly to do many things. Wonderful new behavior, but hard to predict exactly.

Text, images, code on demand

From autocomplete to agents

🕒 9:10 – 9:25 · 15 min

01

Large language model

Guess the next word

Words in, word out. Trained on huge amounts of text.



02

Chatbot

Hold a conversation

Tuned with human feedback to follow instructions.



03

Reasoning model

Work through a problem

Writes hidden steps before it answers.



04

Agentic system

Act in the world

Searches the web, runs code, sends email, edits files.

Each step adds capability, and each step adds new ways to fail.

The good, the bad, and the ugl-AI

🕒 9:10 – 9:25 · 15 min



- Protein structure prediction (AlphaFold)
- Real progress on language tasks



- Rising electricity costs
- Copyright and intellectual property



- Question answering, but it makes things up
- Study aid, but it can block learning
- Companionship, but people can spiral

Fluent, confident, and wrong

🕒 9:10 – 9:25 · 15 min

AR

Can you tell me about the Provost of Tulane University?



As of my knowledge cutoff date of September 2021, the Provost of Tulane University was Dr. Robin Forman. The Provost is the chief academic officer of the university and is responsible for overseeing all academic programs, faculty appointments and promotions, and research initiatives.

Dr. Forman was appointed as Tulane University's 17th Provost in 2010, and he has also served as a professor of political science at the university since 1988. Prior to his appointment as Provost, Dr. Forman served as the Dean of the School of Liberal Arts at Tulane University and was instrumental in leading the school's recovery and rebuilding efforts following Hurricane Katrina in 2005.

As Provost, Dr. Forman has focused on enhancing Tulane's academic programs and research initiatives, as well as promoting interdisciplinary collaboration across the university. He has also worked to expand Tulane's global engagement and community partnerships, with a particular focus on New Orleans and the Gulf Coast region.

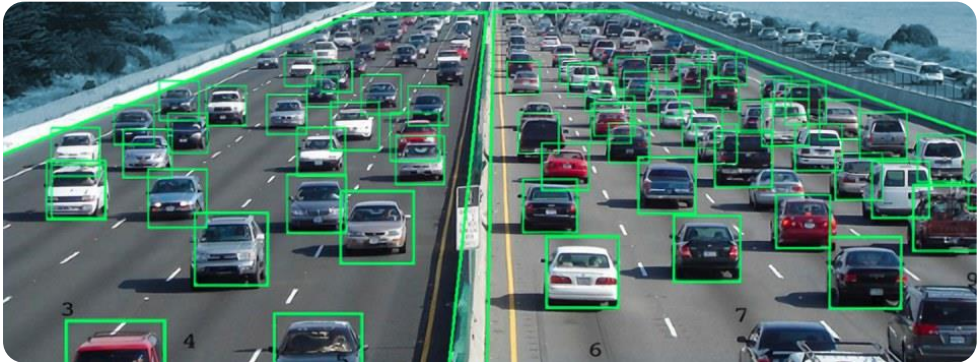
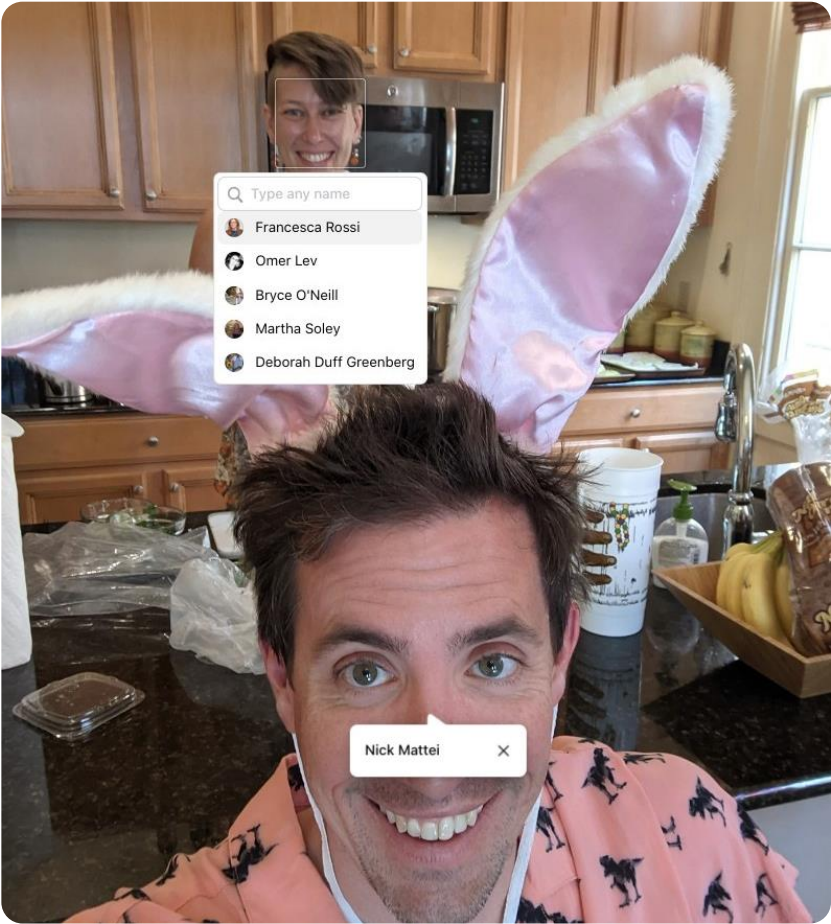
Ask a chatbot about Tulane's provost and you get a detailed, polished biography, with details that are made up.

Often called “hallucination.” Nick avoids the word: it anthropomorphizes a random number generator.

Check every output. Test, test, and test again.

What's at stake?

🕒 9:10 – 9:25 · 15 min



What's at stake, right here in New Orleans

🕒 9:10 – 9:25 · 15 min

EXCLUSIVE

Police secretly monitored New Orleans with facial recognition cameras

Following records requests from The Post, officials paused the first known, widespread live facial recognition program used by police in the United States.

May 19, 2025

🔊 18 min

🔖

📌

💬 635

By [All Winston](#)
Illustrations by Garrett Board
and Alex Cadden
Feb 27, 2018, 2:25 PM CST

🔗

📧

📧

Comments

Palantir has secretly been using New Orleans to test its predictive policing technology

Palantir deployed a predictive policing system in New Orleans that even city council members don't know about

If you buy something from a Verge link, The Verge may earn a commission. [See our disclosure.](#)

npr

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NPR 24 Hour Program Stream

On Air Now

HOURLY NEWS · LISTEN LIVE · MY PLAYLIST

TECHNOLOGY

Live cameras are tracking faces in New Orleans. Who should control them?

DECEMBER 16, 2025 · 4:29 AM ET

HEARD ON MORNING EDITION

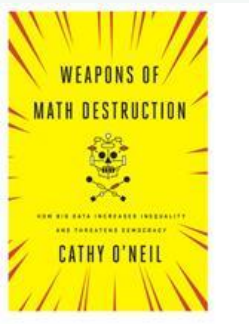
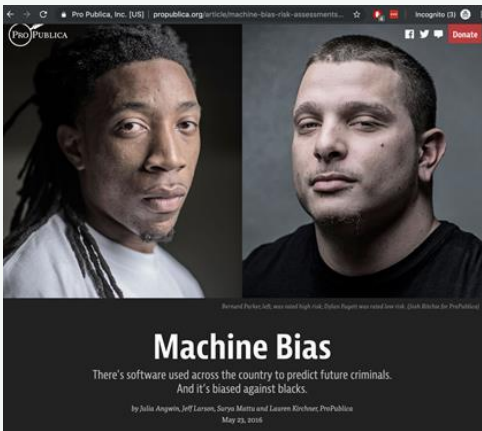
👤

Martin Kaste

Facial recognition and predictive policing were tested on New Orleans residents, often without public knowledge.

Automated decisions of consequence

🕒 9:10 – 9:25 · 15 min



ProPublica, Machine Bias (2016)
O'Neil, Weapons of Math Destruction (2016)

Hiring

A recruiting tool learned to downgrade women's résumés and was scrapped.

Advertising

Searching Black-identifying names surfaced more ads for arrest records.

Lending

Credit models can reproduce historical patterns of exclusion.

Policing and sentencing Risk scores shape bail and sentencing decisions.

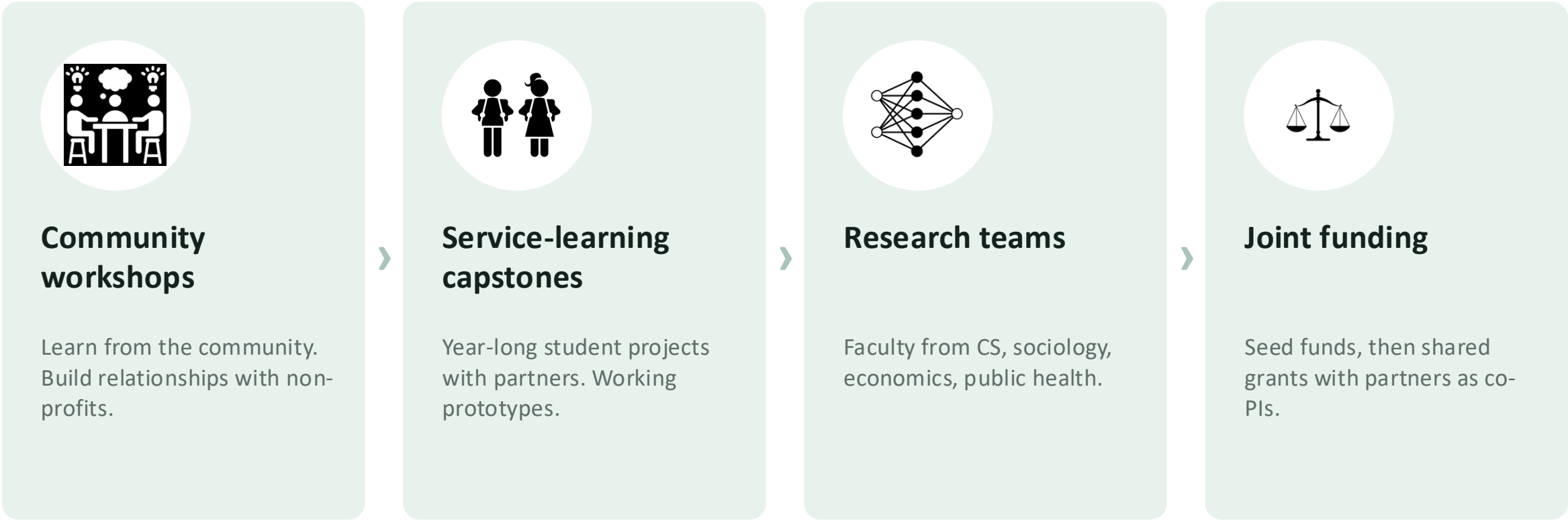


How can academia collaborate outside the ivory tower to solve problems using AI?

Illustration: Melania Bianucci, Ludicomix

A community-led research pipeline

🕒 9:10 – 9:25 · 15 min



Tulane University
Center for Community-Engaged
Artificial Intelligence

sse.tulane.edu/cs/ceai

LLMs for city government

EXAMPLE 1

with Eye on Surveillance

THE CHALLENGE

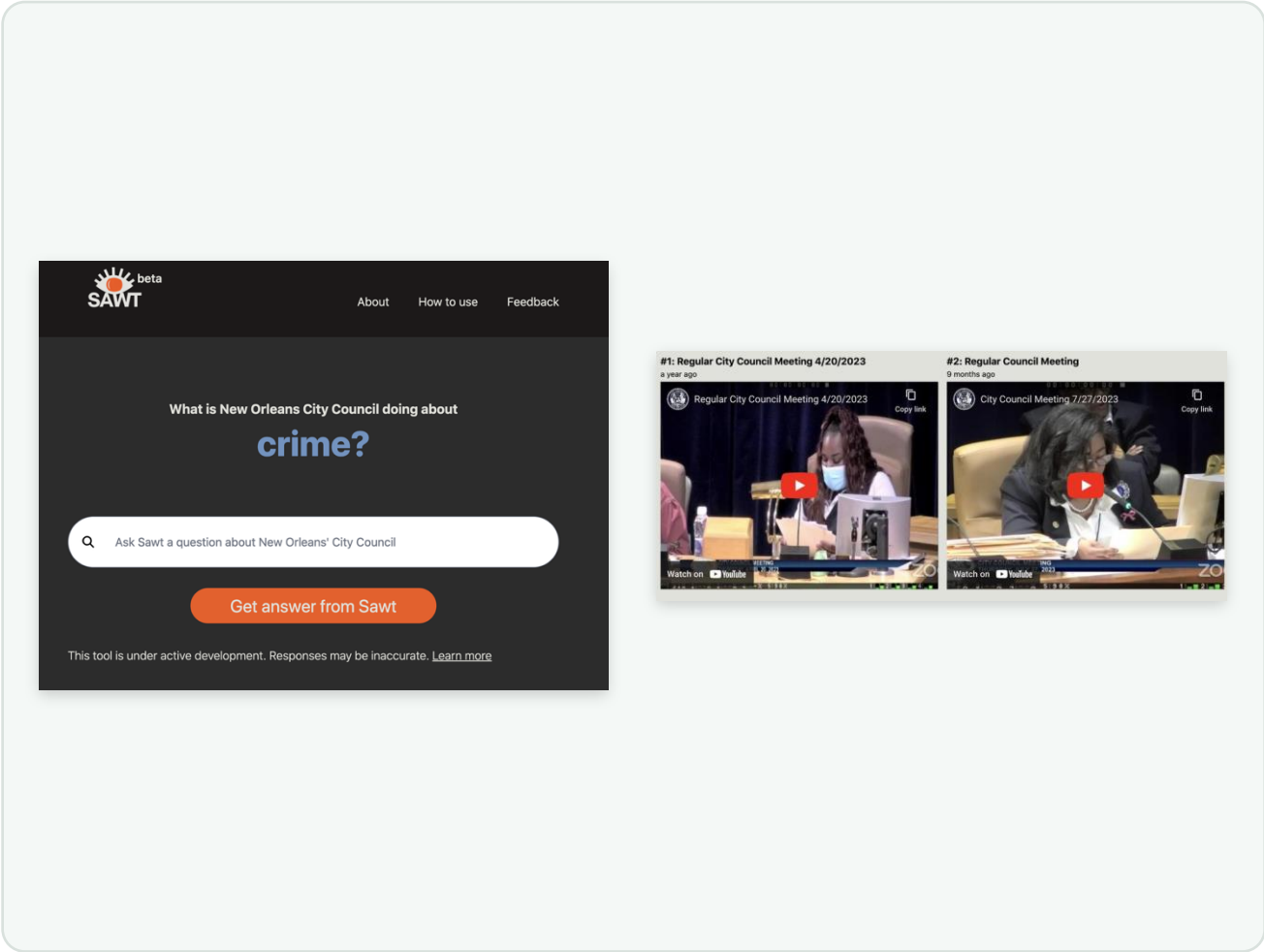
City Council meetings are hours long and hard for residents to search.

WHAT WE BUILT

SAWT: ask a question, get answers drawn from council meeting records.

LEARN MORE

sawt.us



REGGIE: searchable utility meetings

🕒 9:10 – 9:25 · 15 min

EXAMPLE 2

with Entergy

THE CHALLENGE

Public Service Commission meetings: 5 jurisdictions, 5+ hour videos every month.

WHAT WE BUILT

REGGIE, a chatbot that indexes public meetings for question answering.

IMPACT

PSC research went from hours to minutes. Best CS Project, Tulane Capstone Expo.



AI for monitoring criminal court

EXAMPLE 3

with Court Watch NOLA

THE PARTNER

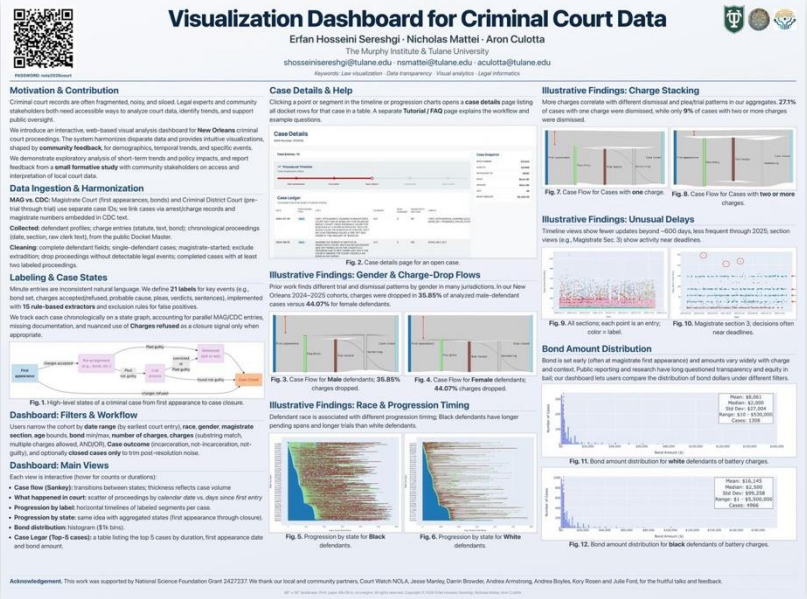
Since 2007, trained volunteers have observed Orleans Parish criminal court.

STARTED SMALL

Student-built data infrastructure for 100K+ cases over 15 years.

GREW INTO

An NSF-funded program: anomaly detection and LLM tools for defendants, victims, and attorneys.



Fairness in microloan discovery

🕒 9:10 – 9:25 · 15 min

EXAMPLE 4

with Kiva.org

THE PARTNER

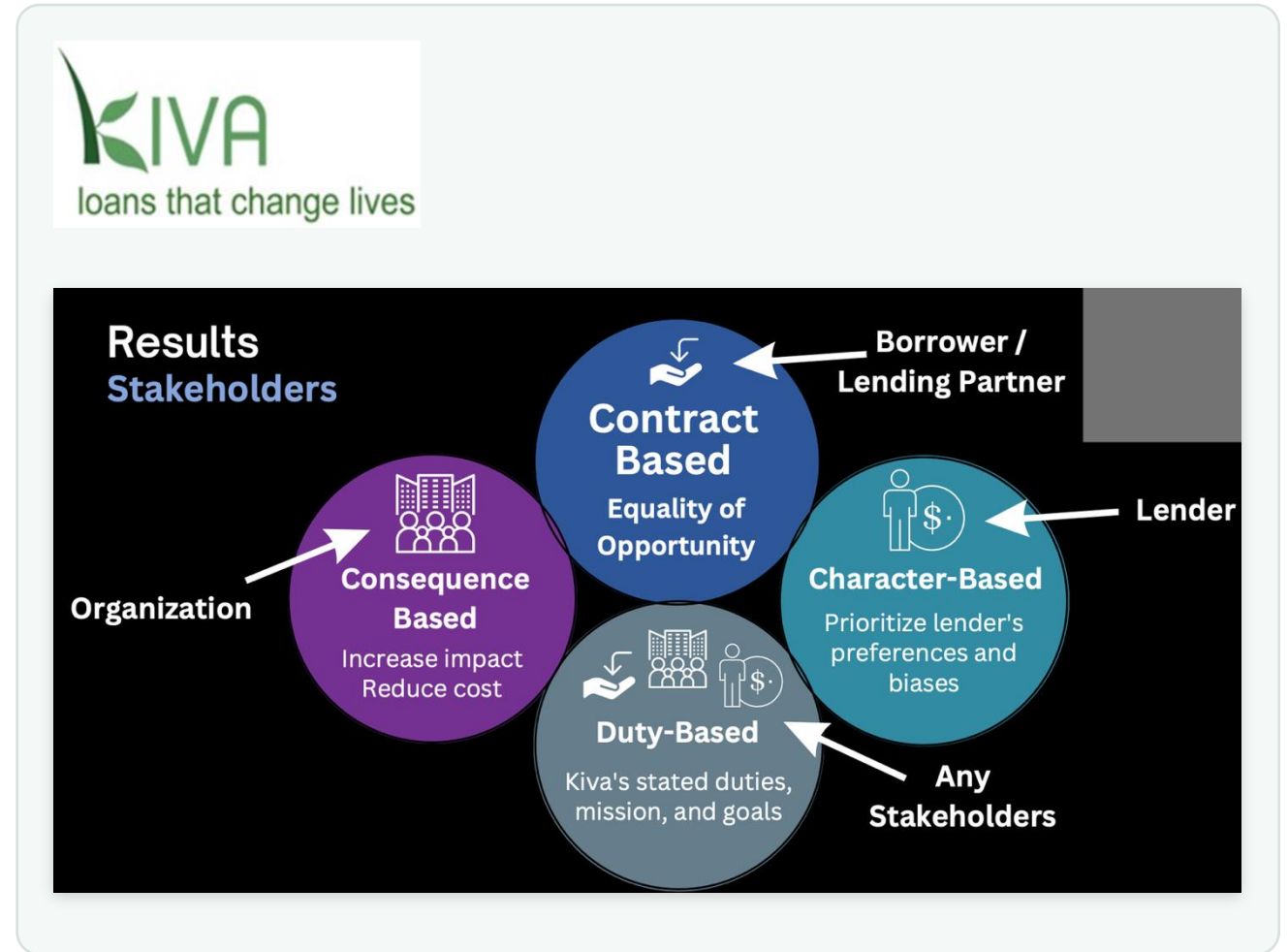
Kiva crowdfunds microloans: \$1.74B to 4.3M borrowers in 77 countries (2022).

THE CHALLENGE

Lender attention is uneven, so some borrowers get overlooked.

WHAT WE STUDIED

Recommendations that balance fairness across borrowers, lenders, and Kiva's mission.



Good practices we've learned

🕒 9:10 – 9:25 · 15 min

1 Start with conversations

Outreach and ideation meetings, with help from your public service office.

2 Start small

Data prep, websites, capstone-sized projects. Early wins build trust.

3 Share the money and the credit

Partners as co-PIs. Budget for their time.

4 Plan for years, not semesters

Hand work off between student cohorts.

5 Listen deeply

Including about when not to use AI.

“Not a pipeline in which communities supply problems that are appropriated by academics.”

Culotta & Mattei, Service Learning for Community-Centered AI Research

PART 2

Designing with, not for

Saad Hassan

HCI, and user-centered, human-centered, and participatory design

What is HCI?

🕒 9:25 – 9:40 · 15 min

“A discipline concerned with the design, evaluation, and implementation of interactive computing systems for human use.”

ACM SIGCHI Curricula for Human-Computer Interaction, 1992

1 Understand people

2 Design with them

3 Build interactive systems

4 Evaluate in real use

DRAWS ON

Computer science

Psychology and cognitive science

Design

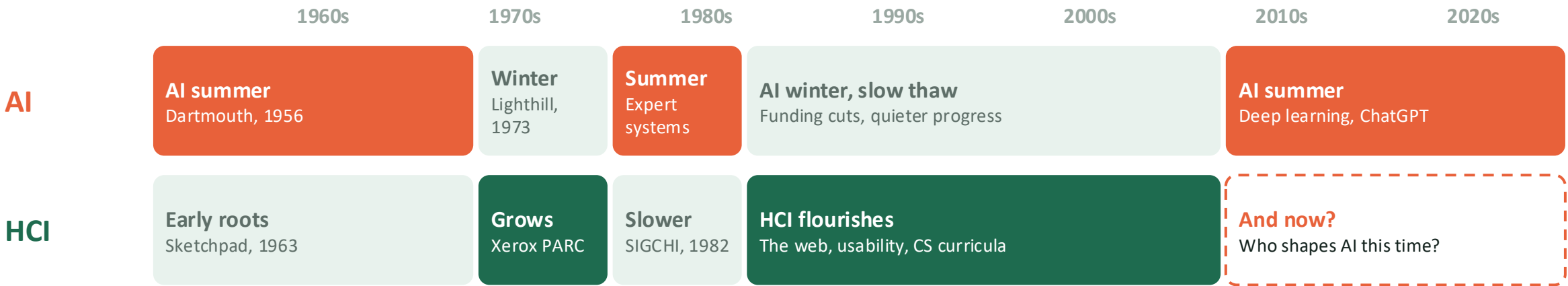
Sociology and anthropology

Disability studies

AI summers, HCI winters, and vice versa

HCI has flourished in AI winters, and moved more slowly when AI was in favor.

Grudin, “AI and HCI: Two Fields Divided by a Common Focus,” AI Magazine, 2009



We are in another AI summer.

This is when HCI and participatory design matter most.

What is design?

🕒 9:25 – 9:40 · 15 min

“The transformation of existing conditions into preferred ones.”

Herbert Simon, 1969

Preferred by whom?

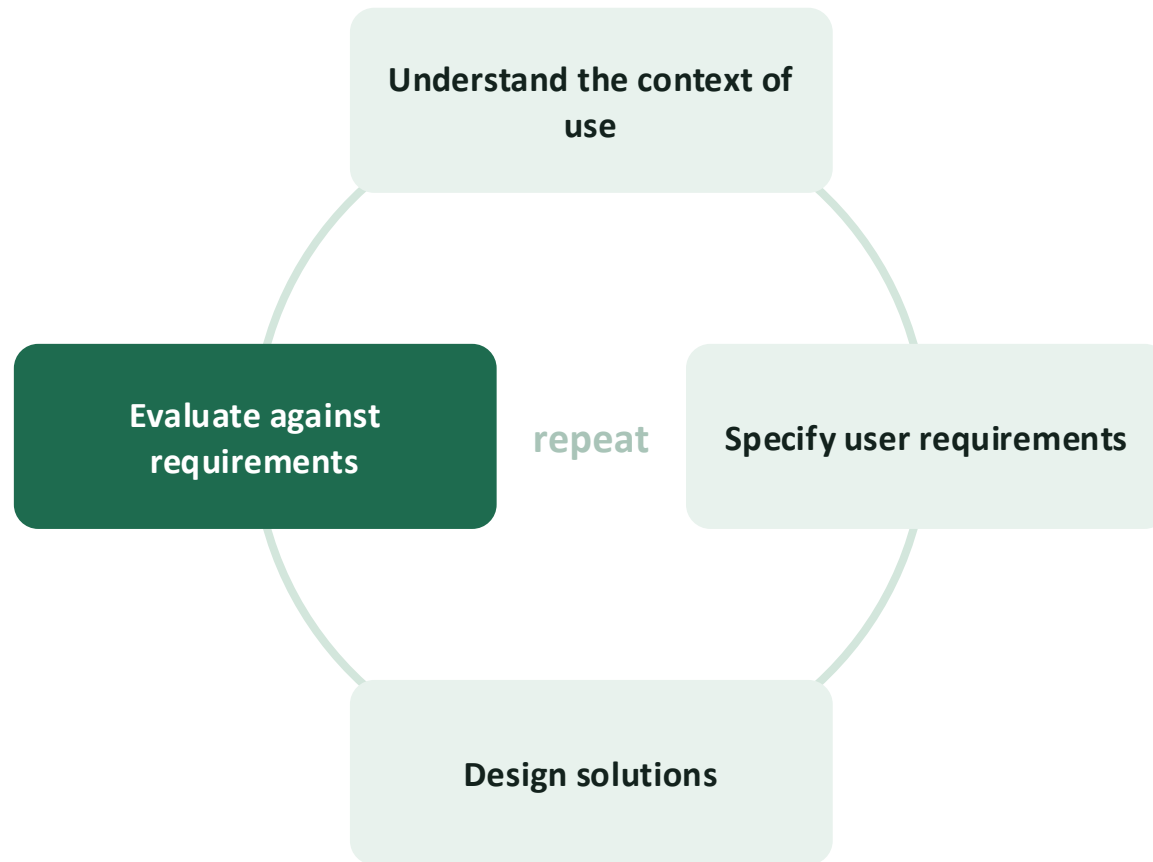
Community problems are wicked problems: requirements conflict or are unknown, and there is no single correct answer (Rittel & Webber, 1973).

Widen the sidewalks, and traffic shifts onto the streets.



User-centered design (UCD)

🕒 9:25 – 9:40 · 15 min



Early and continual focus on users, measurement of real use, and iterative design.

Norman & Draper, 1986

The limitation

Users are studied and tested with, but designers still decide what counts as a requirement. Power stays with the design team.

Human-centered design (HCD)

🕒 9:25 – 9:40 · 15 min

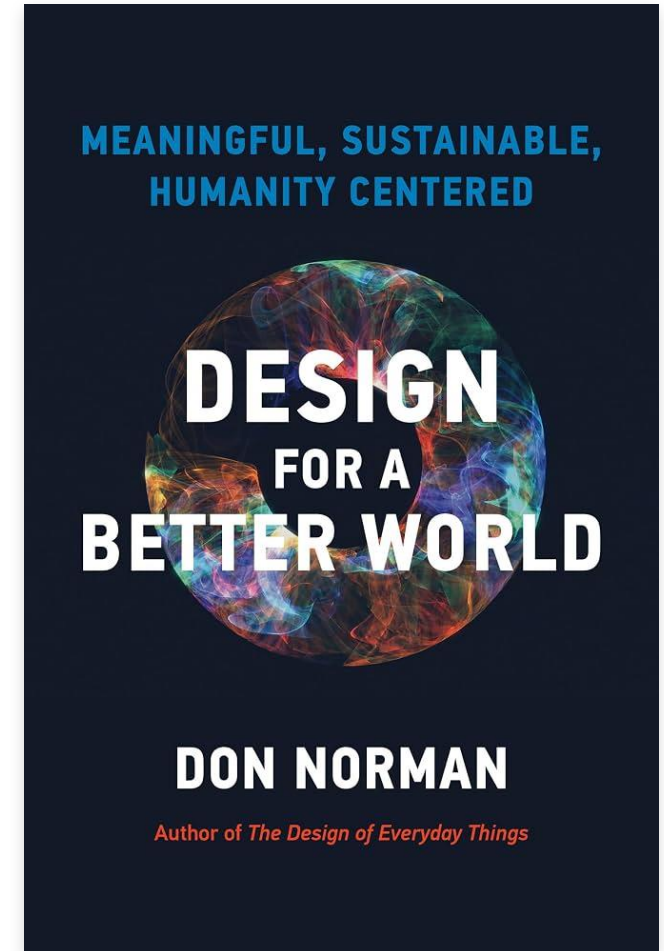
Widens the lens from users to people, communities, and the systems around them.

1 Immersion

2 Solve the root problem

3 Think in systems

4 Iterate, small and often



Two warnings from the research

🕒 9:25 – 9:40 · 15 min

Is empathy enough?

Simulating disability (for example, wearing a blindfold) can lend false confidence and lead to wrong conclusions.

Bennett & Rosner, “The Promise of Empathy: Design, Disability, and Knowing the ‘Other’,” CHI 2019

Is a seat at the table enough?

Bringing people into the room does not mean they have, or see themselves as having, full status and authority.

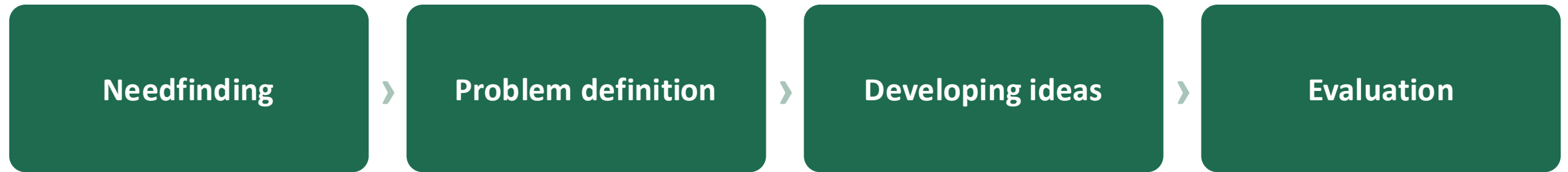
Harrington, Erete & Piper, “Deconstructing Community-Based Collaborative Design,” CSCW 2019

Participatory design (PD)

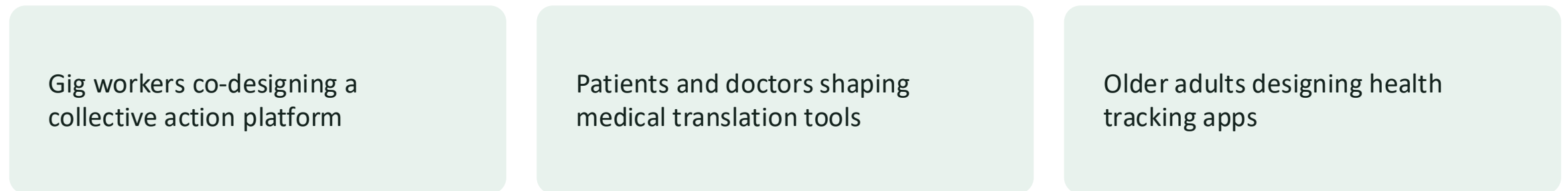
🕒 9:25 – 9:40 · 15 min

De-centers the designer. Stakeholders share decisions in every phase.

Schuler & Namioka, 1993



EXAMPLES



Between Scylla and Charybdis

🕒 9:25 – 9:40 · 15 min



The Odyssey (Universal Pictures, 2025)

Too few voices

“You should have included group X.” A design that misses the mark.

Too many, with no stopping point

Is there an ethical point to stop adding voices and still make decisions?

Where can communities shape AI?

🕒 9:25 – 9:40 · 15 min

1

Problem framing

What should we work on?

2

Data

Whose data, collected how?

3

Design

What does it look like?

4

Evaluation

Who decides it works?

5

Governance

Who deploys and oversees it?

And the question before all of these:

Should AI be used here at all?

Sign language for DHH signers

🕒 9:25 – 9:40 · 15 min

EXAMPLE 1

Deaf-led, with Google and DPAN

PERSONAL ASSISTANTS

Assistants and smart speakers assume speech. We study how Deaf signers want to use them in sign language.

TEXT ENTRY · MOBILEHCI 2023

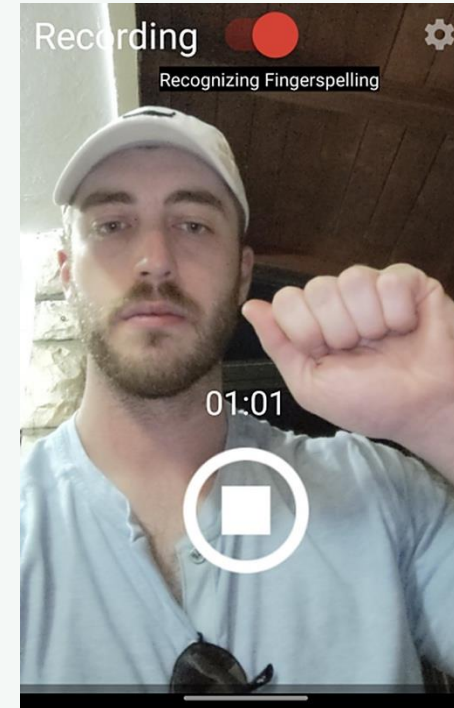
Fingerspelling on a phone camera was faster and more accurate than typing on the keyboard (Tap to Sign).

LARGE-SCALE DATA · CVPR 2025

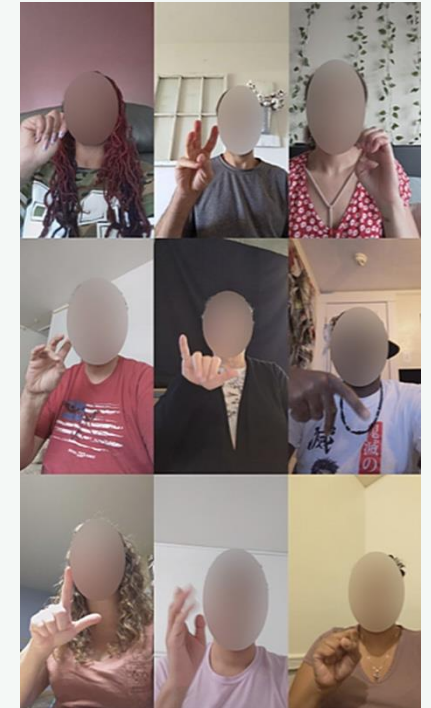
FSboard: a large fingerspelling dataset recorded by Deaf signers on their own phones, with Google and DPAN.

DEAF-LED

Built with Deaf researchers and the Deaf Professional Arts Network (DPAN), from the questions to the data.



Tap to Sign · MobileHCI 2023



FSboard · CVPR 2025

Sign language for learning

EXAMPLE 2

Families and healthcare

FAMILIES

Hearing parents learning to sign with their Deaf and DeafBlind children. PopSign game and dataset, NeurIPS 2023.

HEALTHCARE

Sign vocabulary for body parts, symptoms, and medical terms. ASSETS 2025.

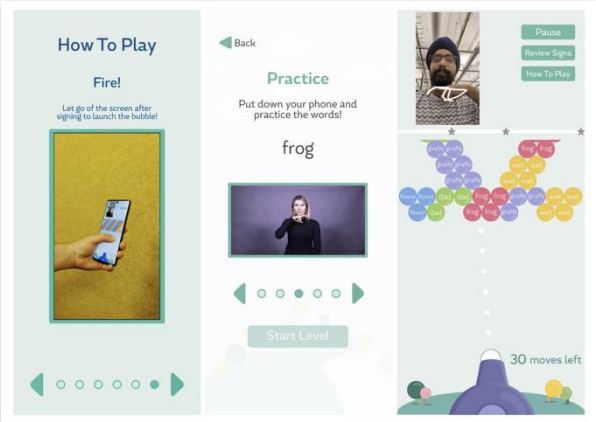
PARTNER

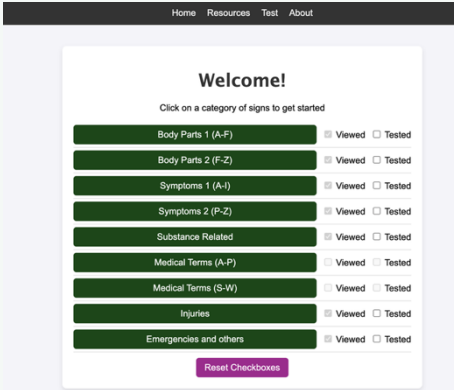
Commission for Deaf & Hard of Hearing

DIDACTIC RECIPROCITY

Designing for people with disabilities, and also for the communities around them: Deaf signers teach, hearing families learn.









Captions that carry emotion

EXAMPLE 3

Deaf & hard-of-hearing viewers

THE CHALLENGE

Captions drop tone of voice: how something is said, not just what.

WHAT WE BUILT

Caption Royale: AI estimates emotion from each spoken word, and the caption’s typography shows it.

PUBLISHED

CHI 2024 (Honorable Mention), ASSETS 2024, CHI 2025

1 · SPEECH

Every word

Aligned to the audio, word by word



>

2 · AI MODEL

Estimates emotion

Valence (unpleasant ↔ pleasant) and arousal (calm ↔ intense)



>

3 · TYPOGRAPHY

Caption shows it

Color carries valence. Weight or size carries arousal.



We **finally** heard back, and **we got it!**

negative

neutral

positive

heavier = more intense



+ HAPTICS

A wristband adds emotion through vibration. Tactile Emotions, CHI 2025.

Civic AI for accessibility barriers

🕒 9:25 – 9:40 · 15 min

EXAMPLE 4

Civic & disability organizations

Dashboards and AI that help civic groups, residents, and local governments see and act on accessibility barriers.

“If I could ask, ‘Give me a list of all ADA sidewalk problems on North State Street between X and Y,’ and it answered accurately and verifiably, that would be game-changing.”

NO ONE SIZE FITS ALL

Civic AI has to adapt to different people, roles, and access needs.

AI summaries with local citations
Answers linked to authoritative records.

Escalate to a human
A visible path to staff when automation falls short.

Mark where AI is used
Plain language and clear onboarding.

NEW NSF AWARD · 2026

Civic AI that connects residents, communities, and local governments.

Your one problem

🕒 9:40 – 10:10 · 30 min

Pick one community engagement challenge where AI could help, hurt, create barriers, or lower them.

- 1

The problem

One sentence
- 2

Who is affected

Partners, students, residents
- 3

Where AI might help

Even a little
- 4

Where AI worries you

Risks, harms, trust
- 5

Who isn't at the table yet

Voices to bring in

○ Circle

the one line you most want to dig into: a hope or a worry.

Share in 1–2 sentences

Your problem, plus what you circled. Sticky note goes on the wall.

9:40 · 15 min to write

9:55 · 15 min to share

Break

15

minutes

Back at 10:25

While you're away

- 1 We group your sticky notes into 2–4 shared themes
- 2 Each theme becomes a breakout starting point
- 3 Add another sticky note if something new comes up

Breakout groups and share-outs

🕒 10:25 – 11:20 · 55 min

Group A
with Nick

Group B
with Saad

Pick the group closest to your problem.

10:25

15 min

Elaborate the problem

Add context on sticky notes. What's really going on?

10:40

15 min

Who else belongs in this conversation?

Partners and community members you could bring in.

10:55

15 min

Limits of AI

Where should human agency stay? Where does human-centered work stand after AI?

11:10

Share out: 5 min per group. One problem, one partner to bring in, one line AI shouldn't cross.

Debrief

🕒 11:20 – 11:30 · 10 min

1 One idea you'll take home

2 One open question

3 What follow-up would help you?



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Thank you!